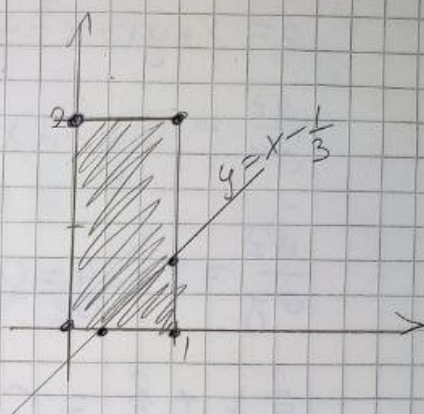


3469

$$\iint_D (x + xy - x^2 - y^2) d\sigma, \quad D: 0 \leq x \leq 1 \\ 0 \leq y \leq 2$$



$$f = (x + xy - x^2 - y^2)$$

Критические экстремумы: $\frac{\partial f}{\partial x} = \frac{\partial f}{\partial y}$

$$\frac{\partial f}{\partial x} = 1 + y - 2x$$

$$\frac{\partial f}{\partial y} = 0 + x - 0 - 2y = x - 2y$$

$$\begin{cases} 1 + y - 2x = x - 2y \\ 1 = 3x - 3y \\ y = x - \frac{1}{3} \end{cases}$$

$$\frac{\partial^2 f}{\partial x^2} = -2 \quad \frac{\partial^2 f}{\partial x \partial y} = 1 \quad \frac{\partial^2 f}{\partial y^2} = -2$$

$$\Delta = \frac{\partial^2 f}{\partial x^2} \cdot \frac{\partial^2 f}{\partial y^2} - \left(\frac{\partial^2 f}{\partial x \partial y} \right)^2 < 0$$

$$-2 \cdot (-2) - 1^2 = 3 > 0, \quad \frac{\partial^2 f}{\partial x^2} = -2 < 0 \Rightarrow \pi. Mo - \pi. max$$

$$\pi. Mo: \begin{cases} \frac{\partial f}{\partial x} = 0 \\ \frac{\partial f}{\partial y} = 0 \end{cases} \Rightarrow \begin{cases} 1 + y - 2x = 0 \\ x - 2y = 0 \end{cases} \Rightarrow \begin{cases} y = 2x - 1 \\ 1 = 3y \end{cases} \Rightarrow Mo\left(\frac{2}{3}, \frac{1}{3}\right)$$

$$f_{max} = f(Mo) = \frac{2}{3} + \frac{2}{3} \cdot \frac{1}{3} - \frac{4}{9} - \frac{1}{9} = \frac{1}{3}$$

По формуле Лагранжа:

$$z = f(x, y) + \lambda \varphi(x, y)$$

②

1) $\varphi = x - 0$:

$$z = x + yx - x^2 - y^2 + \lambda x$$

$$\frac{\partial z}{\partial x} = 1 + y - 2x + \lambda = 0$$

$$\frac{\partial z}{\partial y} = x - 2y = 0$$

$$\begin{cases} y = \frac{1}{3}\lambda + \frac{1}{3} \\ x = \frac{2}{3}\lambda + \frac{2}{3} \end{cases}$$

$$\frac{2}{3}\lambda + \frac{2}{3} = 0 \Rightarrow \lambda = -1 \Rightarrow \begin{cases} x = 0 \\ y = 0 \end{cases}$$

$A(0;0)$

$f(A) = 0$

2) $\varphi = x - 1$:

$$z = x + yx - x^2 - y^2 + \lambda(x-1)$$

$$\frac{\partial z}{\partial x} = 1 + y - 2x + \lambda = 0$$

$$\frac{\partial z}{\partial y} = x - 2y = 0$$

$$\Rightarrow \begin{cases} y = \frac{1}{3}\lambda + \frac{1}{3} \\ x = \frac{2}{3}\lambda + \frac{2}{3} \end{cases}$$

$$\frac{2}{3}\lambda + \frac{2}{3} = 1 \Rightarrow \lambda = \frac{1}{2} \Rightarrow \begin{cases} x = 1 \\ y = \frac{1}{2} \end{cases}$$

$B(1; \frac{1}{2})$

$f(B) = 1 + \frac{1}{2} - 1 - \frac{1}{4} = \frac{1}{4}$

3) $\varphi = y - 0$:

$$z = x + yx - x^2 - y^2 + \lambda(y-0)$$

$$\frac{\partial z}{\partial x} = 1 + y - 2x = 0$$

$$\frac{\partial z}{\partial y} = x - 2y + \lambda = 0$$

$$\Rightarrow \begin{cases} y = \frac{2x+1}{3} \\ x = \frac{\lambda+2}{3} \end{cases}$$

$$\frac{2x+1}{3} = 0 \Rightarrow x = -\frac{1}{2} \Rightarrow \begin{cases} y = 0 \\ x = -\frac{1}{2} \end{cases}$$

$C(-\frac{1}{2}; 0)$

$$f(c) = 2 + 0 - \frac{1}{4} - 0 = \underline{\underline{\frac{7}{4}}}$$

$$4) \varphi = y - 2$$

$$z = x + xy - x^2 - y^2 + \lambda(y - 2)$$

$$\begin{cases} \frac{\partial z}{\partial x} = 1 + y - 2x = 0 \\ \frac{\partial z}{\partial y} = x - 2y + \lambda = 0 \end{cases} \Rightarrow \begin{cases} y = \frac{2}{3}\lambda + \frac{1}{3} \\ x = \frac{1}{3}\lambda + \frac{2}{3} \end{cases}$$

$$\frac{2\lambda + 1}{3} = 2 \Rightarrow \lambda = \frac{5}{2} \Rightarrow \begin{cases} y = 2 \\ x = \frac{5}{3} \end{cases} \in D(\frac{5}{3}; 2)$$

$$f(D) = \frac{5}{3} + \frac{5}{3} \cdot 2 - \frac{25}{9} - 4 = -\frac{16}{9}, \quad \tau D \neq \partial D$$

$$\neq \tau: \underbrace{(0; 0), (1; 0), (0; 2), (1; 2)}_{\tau, \text{ограничивающие } D}, \underbrace{(\frac{1}{3}, 0), (1, \frac{2}{3})}_{\in y = x - \frac{1}{3}}$$

$$f(0; 0) = 0$$

$$f(1; 0) = 1 + 0 - 1 - 0 = 0$$

$$f(0; 2) = 0 + 0 - 0 - 4 = -4 \text{ — мин значение } f, x, y \in D$$

$$f(1; 2) = 1 + 1 \cdot 2 - 1 - 4 = -2$$

$$f(\frac{1}{3}; 0) = \frac{1}{3} + 0 - \frac{1}{9} - 0 = \frac{2}{9}$$

$$f(1, \frac{2}{3}) = 1 + \frac{2}{3} - 1 - \frac{4}{9} = \frac{2}{9}$$

$$f_{\max} = \frac{1}{3} \text{ см ср 1} \quad f_{\min} = -4$$

$$S \cdot f_{\min} \leq \int \leq S \cdot f_{\max}$$

$$S = 2 \cdot 1 = 2 \Rightarrow -4 \cdot 2 \leq I \leq \frac{2}{3} \\ -8 \leq I \leq \frac{2}{3}$$

(4)

3485

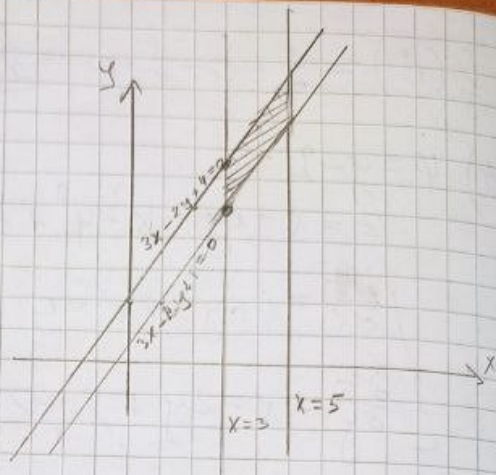
$$\iint_D f(x, y) dx dy = ?$$

Параметры : $x = 3$

$$x = 5$$

$$3x - 2y + 4 = 0$$

$$3x - 2y + 1 = 0$$



$$3x - 2y + 4 = 0 \Leftrightarrow y = \frac{3}{2}x + 2$$

$$3x - 2y + 1 = 0 \Leftrightarrow y = \frac{3}{2}x + \frac{1}{2}$$

$$\iint_D f(x, y) dx dy = \int_3^5 dx \int_{\frac{3}{2}x + \frac{1}{2}}^{\frac{3}{2}x + 2} dy = \int_3^5 dx \cdot \left(\frac{3}{2}x + 2 - \frac{3}{2}x - \frac{1}{2} \right) =$$

$$= \frac{3}{2} \cdot \int_3^5 dx = \frac{3}{2} (5 - 3) = 3$$

3496

$$\begin{cases} y^2 \leq 8x \rightarrow x \geq 0 \\ y \leq 2x \end{cases}$$

$$y \leq 2x$$

$$y + 2 \cdot 2x - 2y \leq 0 \Leftrightarrow y \leq 4x$$

$$\begin{cases} y = 2x \\ y^2 = 8x \end{cases}$$

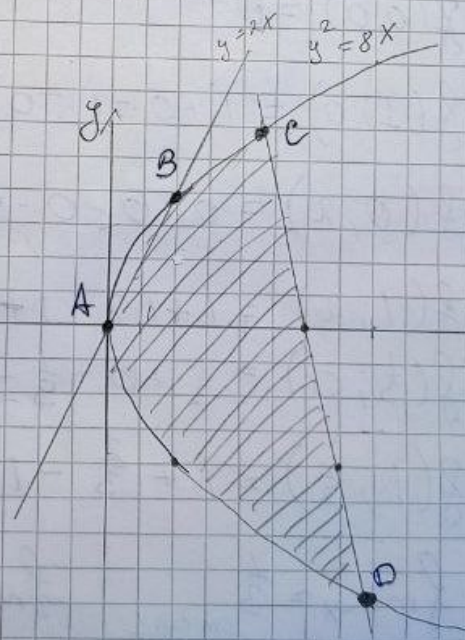
$$\Rightarrow \begin{cases} x = 0 \\ y = 0 \end{cases}$$

$$A(0, 0)$$

$$\begin{cases} x = 2 \\ y = 4 \end{cases}$$

$$B(2, 4)$$

$$y = 4$$



$$\begin{cases} y^2 = 8x \\ y = -4x + 24 \end{cases}$$

$$\Rightarrow \begin{cases} x = \frac{y^2}{8} \\ y + 4 \cdot \frac{y^2}{8} - 24 = 0 \end{cases}$$

$$\Rightarrow \begin{cases} x = \frac{1}{8}y^2 \\ y^2 + 2y + 1 = 49 \end{cases} \Rightarrow$$

$$\begin{cases} x^2 = \frac{y^2}{8} \\ y+1=4 \\ y+1=-7 \end{cases} \rightarrow \begin{cases} y=6 \\ x=\frac{9}{2} \end{cases} C(4,5;6) \\ \begin{cases} y=-8 \\ x=8 \end{cases} D(8;-8)$$

$$\begin{aligned} S &= \int_A^B \int_{-18x}^{2x} (2x - (-\sqrt{8}x)) dx + \int_B^C \int_{-18x}^{\sqrt{8}x} (\sqrt{8}x - (-\sqrt{8}x)) dx + \int_C^D \int_{-2\sqrt{2}x}^{-4x+24} (24-4x - (-\sqrt{8}x)) dx = \\ &= \int_0^2 dx \int_{-18x}^{2x} dy + \int_2^{9/2} dx \int_{-18x}^{\sqrt{8}x} dy + \int_{9/2}^8 dx \int_{-2\sqrt{2}x}^{-4x+24} dy = \\ &= \int_0^2 (2x + 18x) dx + \int_2^{9/2} (2\sqrt{8}x) dx + \int_{9/2}^8 (-4x + \sqrt{8}x + 24) dx = \\ &= \left(2\frac{x^2}{2} + 18\frac{x^2}{2} \right) \Big|_0^2 + \left(2\sqrt{8} \cdot \frac{x^2}{2} \right) \Big|_2^{9/2} + \left(\frac{x^2}{2} (\sqrt{2} - 4) + 24x \right) \Big|_{9/2}^8 = \\ &= 4 + 4\sqrt{2} + \frac{81}{4} \cdot 2\sqrt{2} - 4 \cdot 2\sqrt{2} + 64 \cdot (\sqrt{2} - 2) + 24 \cdot 8 - \frac{81}{4} (\sqrt{2} - 2) - 24 \cdot \frac{9}{2} = \\ &= 4 + 24 \cdot 8 - 64 \cdot 2 + \frac{81}{4} \cdot 2 - 24 \cdot \frac{9}{2} + \sqrt{2} \left(\frac{81}{2} - 8 + 64 - \frac{81}{4} \right) = \\ &= 1/2 + 76\frac{1}{4} \cdot \sqrt{2} \end{aligned}$$

Ответ: $\int_0^2 dx \int_{-18x}^{2x} f(x,y) dy + \int_2^{9/2} dx \int_{-18x}^{\sqrt{8}x} f(x,y) dy + \int_{9/2}^8 dx \int_{-2\sqrt{2}x}^{-4x+24} f(x,y) dy$

При $f(x,y) = 1$ $S = \frac{1}{2} + 76\frac{1}{4} \cdot \sqrt{2} \approx 108,33$

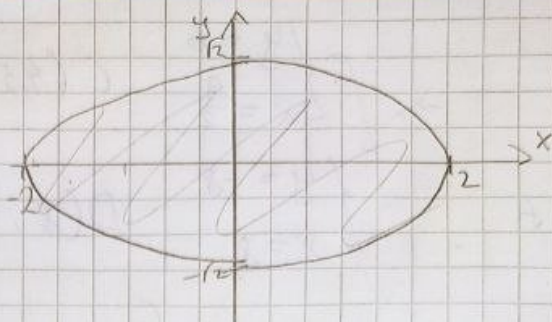
3501

$$\int_{-2}^2 dx \int_{-\frac{1}{2}\sqrt{4-x^2}}^{\frac{1}{2}\sqrt{4-x^2}} f(x,y) dy \quad \text{изменить порядок интегриров}$$

$x \in [-2, 2]$

$$-\frac{1}{2}\sqrt{4-x^2} \leq y \leq \frac{1}{2}\sqrt{4-x^2} \Rightarrow y^2 \leq \frac{1}{2}(4-x^2)$$

⑥ $2y^2 + x^2 \leq 4$
 $\frac{x^2}{4} + \frac{y^2}{2} = 1$ - ellipse



$y \in [-\sqrt{2}; \sqrt{2}]$

$x^2 \leq 4 - 2y^2 \Rightarrow -\sqrt{4 - 2y^2} \leq x \leq \sqrt{4 - 2y^2}$

$\int_{-2}^2 dx \int_{-\frac{1}{\sqrt{2}}\sqrt{4-x^2}}^{\frac{1}{\sqrt{2}}\sqrt{4-x^2}} f(x,y) dy = \int_{-\sqrt{2}}^{\sqrt{2}} dy \int_{-\sqrt{4-2y^2}}^{\sqrt{4-2y^2}} f(x,y) dx$

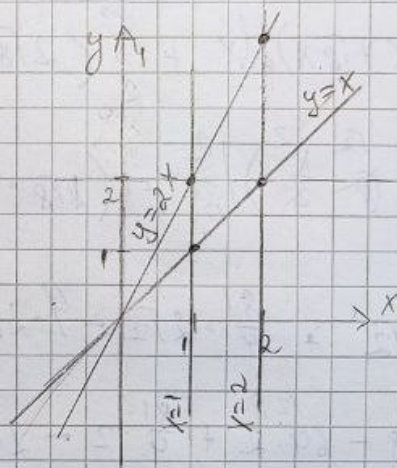
$\int_1^2 dx \int_x^{2x} f(x,y) dy$

3502

$x \in [1; 2]$

$x \leq y \leq 2x \Rightarrow 1 \leq y \leq 4$

$\begin{cases} x=y \\ x=\frac{1}{2}y \end{cases}$



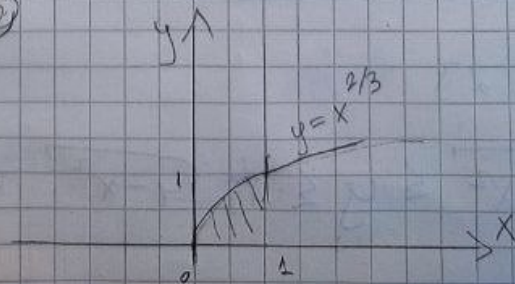
Или $1 \leq y \leq 2$ $1 \leq x \leq y$, или $2 \leq y \leq 4$ $\frac{1}{2}y \leq x \leq 2$

$\int_1^2 dx \int_x^{2x} f(x,y) dy = \int_1^2 dy \int_1^y f(x,y) dx + \int_2^4 dy \int_{\frac{1}{2}y}^2 f(x,y) dx$

3504 (3)

$\int_0^1 dx \int_0^{x^{2/3}} f(x,y) dy + \int_1^2 dx \int_0^{1-\sqrt{4x^2-3}} f(x,y) dy$

* ① $x \in [0; 1]$
 $0 \leq y \leq x^{2/3}$



$$1 \leq x \leq y^{\frac{3}{2}}, \quad y \in [0; 1]$$

$$\int_0^1 dy \int_{y^{3/2}}^1 f(x, y) dx$$

$$\text{4) } x \in [1; 2]$$

$$0 \leq y \leq 1 - \sqrt{4x - x^2 - 3} \Rightarrow x^2 - 4x + 4 \leq 1 - \text{нижняя половина}$$

$$y = 1 - \sqrt{4x - x^2 - 3}$$

$$(y-1)^2 = -x^2 + 4x - 3$$

$$(y-1)^2 + (x-2)^2 = 1$$

$$x-2 = \pm \sqrt{1 - (y-1)^2}$$

$$x = 2 + \sqrt{1 - y^2 + 2y - 1} - \text{правая половина}$$

$$x = 2 - \sqrt{2y - y^2} - \text{левая половина}$$

$$x = 2 - \sqrt{2y - y^2}$$

$$0 \leq y \leq 1 \quad \text{и} \quad 1 \leq x \leq 2 - \sqrt{2y - y^2}$$

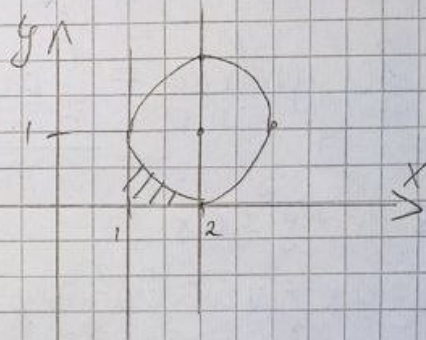
$$\int_0^1 dy \int_{2 - \sqrt{2y - y^2}}^{2 - \sqrt{2y - y^2}} f(x, y) dx$$

$$\text{Таким образом} \quad \int_0^1 dx \int_0^{x^{2/3}} f(x, y) dy + \int_1^2 dx \int_0^{1 - \sqrt{4x - x^2 - 3}} f(x, y) dy =$$

$$= \int_0^1 dy \int_{y^{3/2}}^1 f(x, y) dx + \int_0^1 dy \int_{2 - \sqrt{2y - y^2}}^{2 - \sqrt{2y - y^2}} f(x, y) dx =$$

$$= \int_0^1 \left(\int_{y^{3/2}}^1 f(x, y) dx + \int_{2 - \sqrt{2y - y^2}}^{2 - \sqrt{2y - y^2}} f(x, y) dx \right) dy =$$

$$= \int_0^1 dy \int_{y^{3/2}}^{2 - \sqrt{2y - y^2}} f(x, y) dx$$



8)

3525 К полярным координатам

$$1) x^2 + y^2 \leq R^2 \rightarrow D$$

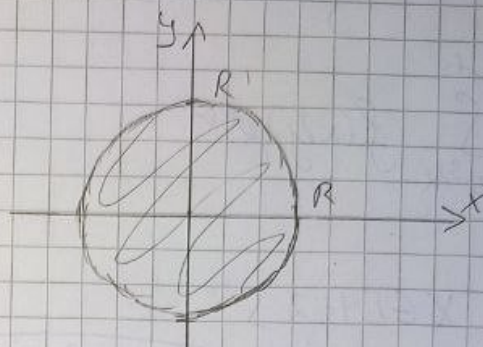
$$0 \leq \varphi \leq 2\pi$$

$$x = r \cos \varphi$$

$$y = r \sin \varphi$$

$$r^2 = x^2 + y^2 \Rightarrow r \leq R$$

$$\iint_D f(x, y) dx dy = \int_0^{2\pi} d\varphi \int_0^R f(r \cos \varphi, r \sin \varphi) r dr$$



$$2) D: x^2 + y^2 \leq ax$$

$$x^2 - ax + \frac{1}{4}a^2 - \frac{1}{4}a^2 + y^2 \leq 0$$

$$\left(x - \frac{a}{2}\right)^2 + y^2 \leq \left(\frac{a}{2}\right)^2$$

$$0 \leq x \leq a, \quad x = r \cos \varphi, \quad y = r \sin \varphi$$

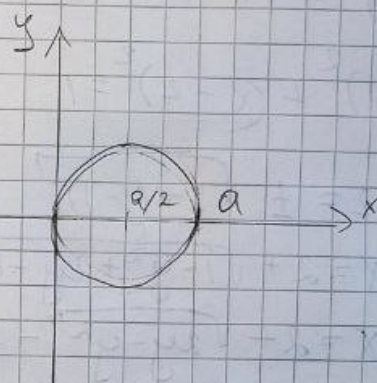
$$r \cos \varphi \geq 0$$

$$\cos \varphi \geq 0 \Rightarrow \varphi \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

$$r^2 \cos \varphi + r^2 \sin^2 \varphi \leq a \cdot r \cos \varphi \quad (r > 0)$$

$$r \leq a \cos \varphi$$

$$\iint_D f(x, y) dx dy = \int_{-\pi/2}^{\pi/2} d\varphi \int_0^{a \cos \varphi} f(r \cos \varphi, r \sin \varphi) r dr$$



$$3) D: x^2 + y^2 \leq by$$

$$\text{аналогично получаем } x^2 + \left(y - \frac{b}{2}\right)^2 \leq \left(\frac{b}{2}\right)^2$$

$$x = r \cos \varphi$$

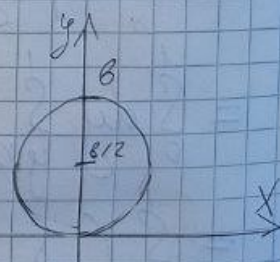
$$r^2 \leq b r \sin \varphi$$

$$y = r \sin \varphi$$

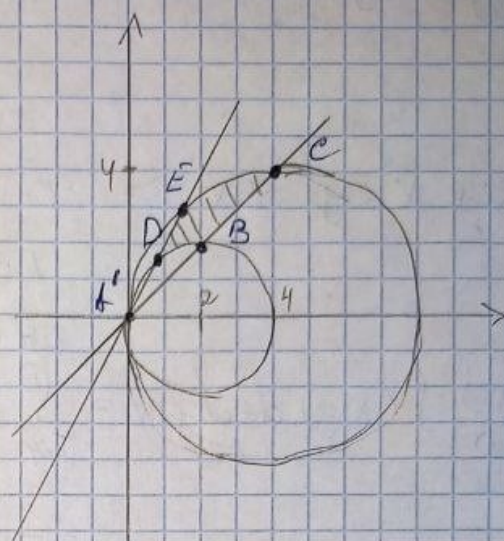
$$b \sin \varphi \geq r \geq 0$$

$$\varphi \in [0; \pi]$$

$$\iint_D f(x, y) dx dy = \int_0^{\pi} d\varphi \int_0^{b \sin \varphi} f(r \cos \varphi, r \sin \varphi) r dr$$



$$D: \begin{cases} x^2 + y^2 = 4x \\ x^2 + y^2 = 8x \\ y = x \\ y = 2x \end{cases}$$



$$D: \begin{cases} (x-2)^2 + y^2 = 2^2 \\ (x-4)^2 + y^2 = 4^2 \\ y = x \\ y = 2x \end{cases}$$

$$\begin{cases} (x-2)^2 + y^2 = 2^2 \\ y = x \end{cases}$$

$$\Rightarrow \begin{cases} y = x \\ x^2 + x^2 = 4x \end{cases}$$

$$\Rightarrow \begin{cases} x=0 \\ y=0 \\ x=2 \\ y=2 \end{cases} \quad \begin{matrix} A(0;0) \\ B(2;2) \end{matrix}$$

$$\begin{cases} x^2 + y^2 = 8x \\ y = x \end{cases}$$

$$\Rightarrow \begin{cases} x^2 + x^2 = 8x \\ y = x \end{cases}$$

$$\Rightarrow \begin{cases} x=0 \\ y=0 \\ x=4 \\ y=4 \end{cases} \quad C(4;4)$$

$$\begin{cases} x^2 + y^2 = 4x \\ y = 2x \end{cases}$$

$$\Rightarrow \begin{cases} y = 2x \\ x^2 + 4x^2 = 4x \end{cases}$$

$$\Rightarrow \begin{cases} x=0 \\ y=0 \\ x=\frac{4}{5} \\ y=\frac{8}{5} \end{cases} \quad D(\frac{4}{5}; \frac{8}{5})$$

$$\begin{cases} x^2 + y^2 = 8x \\ y = 2x \end{cases}$$

$$\Rightarrow \begin{cases} x=0 \\ y=0 \\ x=\frac{8}{5} \\ y=\frac{16}{5} \end{cases} \quad E(\frac{8}{5}; \frac{16}{5})$$

$$\text{wg } y=x \Rightarrow r \sin \varphi = r \cos \varphi \Rightarrow \varphi = \frac{\pi}{4} \text{ und } \frac{5\pi}{4}$$

$$\text{wg } y=2x \Rightarrow r \sin \varphi = 2r \cos \varphi \Rightarrow \tan \varphi = 2 \Rightarrow \varphi = \arctan 2$$

$$\varphi \in [\frac{\pi}{4}; \arctan 2]$$

$$10) \text{ из } x^2 + y^2 = 4x \Rightarrow r^2 \cos^2 \varphi + r^2 \sin^2 \varphi = 4 \cdot r \cos \varphi$$

$$r^2 = 4r \cos \varphi \quad |r > 0$$

$$r = 4 \cos \varphi$$

$$\text{из } x^2 + y^2 = 8x \Rightarrow r^2 = 8r \cos \varphi \Rightarrow r = 8 \cos \varphi$$

$$r \in [4 \cos \varphi, 8 \cos \varphi]$$

$$\iint_D f(x, y) dx dy = \int_{\pi/4}^{\arctan 2} d\varphi \int_{4 \cos \varphi}^{8 \cos \varphi} f(r \cos \varphi, r \sin \varphi) r dr$$

3536

Вычислить с помощью перехода к полярным координатам

$$\int_0^R dx \int_0^{\sqrt{R^2 - x^2}} \ln(1 + x^2 + y^2) dy \quad \ominus$$

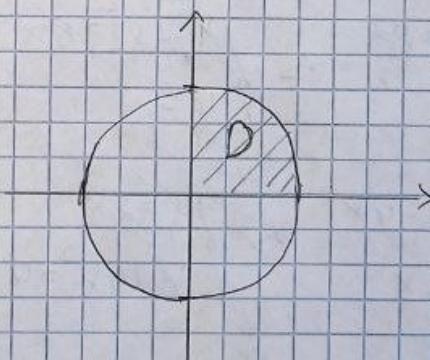
$$0 \leq x \leq R \Rightarrow 0 \leq r \cos \varphi \leq R$$

$$0 \leq y \leq \sqrt{R^2 - x^2} \Rightarrow 0 \leq r \sin \varphi \leq \sqrt{R^2 - r^2 \cos^2 \varphi} \Rightarrow \begin{cases} \cos \varphi \geq 0 \\ \sin \varphi \geq 0 \end{cases} \Rightarrow$$

$$\Rightarrow \varphi \in [0; \pi/2]$$

$$y^2 \leq R^2 - x^2 \Rightarrow y^2 + x^2 \leq R^2$$

$$\begin{aligned} r^2 &\leq R^2 \\ r &\leq R \end{aligned}$$



$$\ominus \int_0^{\pi/2} d\varphi \int_0^R \ln(1 + r^2 \cos^2 \varphi + r^2 \sin^2 \varphi) r dr \quad \equiv$$

$$\text{Вычисляем } \int \ln(1 + r^2) r dr.$$

$$\int \ln(1 + r^2) r dr = \frac{1}{2} \int \ln(1 + r^2) d(r^2 + 1) = \left[\frac{t = r^2 + 1}{d(r^2 + 1) = dt} \right] =$$

$$= \frac{1}{2} \int \ln t dt = \frac{1}{2} [t \ln t - \int t \cdot \frac{1}{t} dt] = \frac{1}{2} (t \ln t - \int dt) =$$

$$= \frac{1}{2} (t \ln t - t).$$

$$\equiv \int_0^{\pi/2} d\varphi \cdot \frac{1}{2} \left((r^2 + 1) \ln(r^2 + 1) - (r^2 + 1) \right) \Big|_0^R = \frac{1}{2} \int_0^{\pi/2} d\varphi \cdot (R^2 \ln(R^2 + 1) +$$

$$+ \ln(R^2+1) - R^2 - 1 - (0 \cdot \ln(0+1) + \ln(0+1) - 0 - 1)] = \quad (11)$$

$$= \int_0^{\pi/2} \frac{1}{2} d\varphi \cdot ((R^2+1) \ln(R^2+1) - R^2) = \frac{\pi}{2} \cdot \frac{1}{2} ((R^2+1) \ln(R^2+1) - R^2) - R^2$$

$$= 0 \cdot \frac{1}{2} ((R^2+1) \ln(R^2+1) - R^2) = \frac{\pi}{4} ((R^2+1) \ln(R^2+1) - R^2)$$

3604

S-2

$$(x^2 + y^2)^2 = 2a^2(x^2 - y^2)$$

$$\begin{cases} x = r \cos \varphi \\ y = r \sin \varphi \end{cases} \Rightarrow (r^2 \cos^2 \varphi + r^2 \sin^2 \varphi)^2 = 2a^2(r^2 \cos^2 \varphi - r^2 \sin^2 \varphi)$$

$$r^4 = 2a^2 r^2 \cos 2\varphi \quad | r \neq 0$$

$$r^2 = 2a^2 \cos 2\varphi^*$$

$$\text{r.K. } 2a^2 \geq 0 \text{ u } r^2 \geq 0 \Rightarrow \cos 2\varphi \geq 0 \Rightarrow -\frac{\pi}{2} \leq 2\varphi + 2\pi n \leq \frac{\pi}{2}$$

$$-\frac{\pi}{4} + \pi n \leq \varphi \leq \frac{\pi}{4} + \pi n$$



$$\varphi \in \left[-\frac{\pi}{4}, \frac{\pi}{4}\right] \cup \left[\frac{3\pi}{4}, \frac{5\pi}{4}\right]$$

$$* \Rightarrow r = \pm a \sqrt{2 \cos 2\varphi}$$



$$S = 4 S_1 = 4 \cdot \iint_{S_1} r dr d\varphi \quad (12)$$

$$S_1: \begin{cases} 0 \leq r \leq a \sqrt{2 \cos 2\varphi} \\ 0 \leq \varphi \leq \frac{\pi}{4} \end{cases}$$

$$\begin{aligned} (12) \quad 4 \cdot \int_0^{\pi/4} d\varphi \int_0^{a \sqrt{2 \cos 2\varphi}} \frac{1}{2} \cdot r dr &= 4 \cdot \int_0^{\pi/4} d\varphi \frac{1}{2} \cdot (a \sqrt{2 \cos 2\varphi})^2 = 2 \int_0^{\pi/4} a^2 \cdot 2 \cdot \\ \cos 2\varphi d\varphi &= 2a^2 \int_0^{\pi/4} \cos 2\varphi d2\varphi = 2a^2 (\sin 2\varphi) \Big|_0^{\pi/4} = 2a^2 (\sin \frac{\pi}{2} - \\ - \sin 0) &= \underline{2a^2} \end{aligned}$$

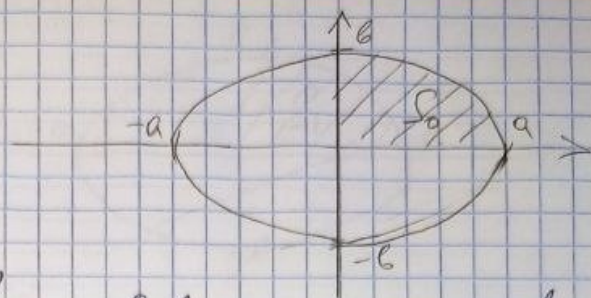
3599

$$D: \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \Rightarrow a, b > 0$$

$$(12) \quad y = \pm \frac{b}{a} \sqrt{a^2 - x^2}$$

$$S_0: 0 \leq x \leq a$$

$$0 \leq y \leq \frac{b}{a} \sqrt{a^2 - x^2}$$



$$S = 4S_0 = 4 \cdot \int_0^a dx \int_0^{\frac{b}{a} \sqrt{a^2 - x^2}} dy = 4 \cdot \int_0^a \frac{b}{a} \sqrt{a^2 - x^2} dx = 4 \frac{b}{a} \cdot$$

$$\int_0^a \sqrt{a^2 - x^2} dx \quad (\Rightarrow)$$

Benennung $\int \sqrt{a^2 - x^2} dx$ Barrena

$$t_1 = \frac{\pi}{2} : a = a \sin t$$

$$t_0 = 0 : 0 = 0 \sin t$$

$$x = a \sin t$$

$$dx = a \cos t dt$$

$$\int \sqrt{a^2 - x^2} dx = \int \sqrt{a^2 - a^2 \sin^2 t} a \cos t dt = a^2 \int \cos^2 t dt =$$

$$= a^2 \int \frac{1 + \cos 2t}{2} dt = \frac{a^2}{4} \int (1 + \cos 2t) d2t = \frac{a^2}{4} \cdot \frac{2t}{1} + \frac{a^2}{4} \cdot \sin 2t =$$

$$= \frac{a^2}{2} \cdot t + \frac{a^2}{4} \cdot \sin 2t$$

$$\int_0^a \sqrt{a^2 - x^2} dx = \left(\frac{a^2}{2} t + \frac{a^2}{4} \sin 2t \right) \Big|_0^{\frac{\pi}{2}} = \frac{a^2}{2} \left(\frac{\pi}{2} + 2 \cdot \sin \pi \right) =$$

$$= \frac{a^2 \pi}{4}$$

$$(\Rightarrow) 4 \frac{b}{a} \cdot \frac{a^2 \pi}{4} = ab\pi$$